THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 8: **Heredity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A heterozygous tall plant (Tt) is crossed with a dwarf plant (tt). The fraction of dwarf plants in the offspring is 1

(a) 3/4 **(b)** 1/4 **(c)** 1/2 **(d)** 0

2. The sex of a baby is determined at the time of 1

(a) puberty **(b)** fertilisation **(c)** birth **(d)** implantation

3. A trait that is controlled by a recessive allele appears in a plant when 1

(a) both alleles are recessive **(b)** no allele is present
(c) both are dominant **(d)** one allele is recessive

4. Two tall plants produce 75 tall and 25 dwarf plants. The parents were most likely 1

(a) both Tt **(b)** both TT **(c)** TT and Tt **(d)** Tt and tt

5. A couple already has two daughters. The probability that their third child is a boy is 1

(a) 1/2 **(b)** 0 **(c)** 1 **(d)** 1/4

6. A bacterium reproducing asexually will show variation because of 1

(a) crossing of two parents **(b)** gamete fusion
(c) minor DNA copying errors **(d)** meiosis in parents

7. Out of 160 F2 plants in a monohybrid cross, the expected number of dwarf plants is 1

(a) 80 **(b)** 120 **(c)** 40 **(d)** 20

8. Which of the following statements about sex determination is correct? 1

(a) Egg decides the sex **(b)** Mother's age decides
(c) Both eggs and sperm always give girls **(d)** Sperm decides the sex

9. Plants of a clone are grown in different soils and show different heights. This difference is 1

(a) a mutation of germ cells **(b)** an acquired trait due to environment
(c) inherited **(d)** dominant

10. Which of the following genotypes will produce only one type of gamete? 1

(a) TT **(b)** Tt **(c)** Both TT and Tt **(d)** None

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. In pea plants, tall (T) is dominant. A tall plant is crossed with a dwarf plant and 50% dwarf offspring result. Find the genotype of the tall parent. 2

12. If a plant with genotype Rr produces 400 seeds by self-fertilisation, how many wrinkled seeds are expected? 2

13. What fraction of the F2 progeny in a dihybrid cross is round (R_)? 2

14. Why does a dwarf trait reappear in the F2 generation though absent in the F1? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A tall pea plant is crossed with a dwarf and the F1 are all tall. When the F1 are self-pollinated, 300 plants result. Find the numbers of tall and dwarf plants and the genotypes. 3

16. Calculate the number of seeds of each phenotype expected in the F2 of a dihybrid cross, if 800 seeds are obtained. 3

17. What would the F1 look like if Mendel had crossed a pure round yellow plant with a pure wrinkled green plant, according to blending theory? Why was blending theory rejected? 3

OR

Show with a Punnett square the cross Rr x Rr for seed shape and give the genotypic and phenotypic ratio. 3

18. What is the role of environment in determining traits? Give an example for plants and humans. 3

OR

A pea plant (Tt) is crossed with another pea plant (Tt). Of 20 offspring, how many will be tall (approx.)? Explain the reasoning behind approximate values. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. In pea plants, round seed (R) is dominant to wrinkled (r), and yellow (Y) to green (y). A plant RrYy is self-pollinated. Make a Punnett square outline: list gametes, give the F2 phenotypes and the ratio. 5

OR

Explain the terms: genotype, phenotype, dominant, recessive and allele, with the help of the cross Tt x Tt. 5

20. A pure tall plant with round seeds is crossed with a dwarf plant with wrinkled seeds. Give the genotype of F1 and the number of each phenotype in F2 out of 160 plants. 5

OR

In humans, free earlobe (F) is dominant over attached (f). A man with attached earlobes marries a woman with free earlobes whose mother had attached earlobes. Predict the children. 5

END OF QUESTION PAPER — **Total: 40 Marks**